Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “latent heat” to explain your model or result. Sentence starters are optional.

phase change: a change between states such as solid and liquid

latent heat: energy per mass needed for a specified phase change

temperature: a measure of thermal state; it is not an energy amount

01 Bring back an earlier idea

A sample warms by 3°C after 500 J is transferred into it.
Explain why the temperature rise and heat transfer need different units.

Word help: heat = energy transferred because of a temperature difference

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: Q = mL; Q = mcΔT.
Use mL during a phase change at constant temperature, and mcΔT outside that interval. L is latent heat.

Worked reasoning: Melting 2 kg of a substance with latent heat 100 J/kg requires 200 J. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting.
Mark the plateau.

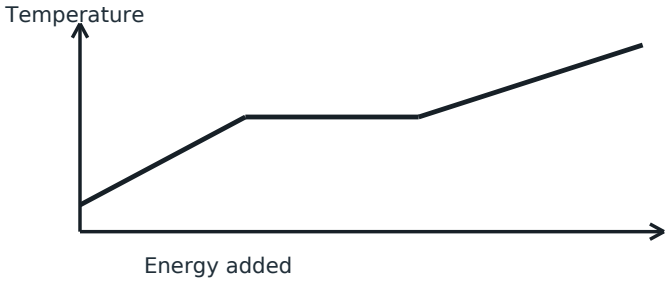
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

On the heating curve, identify the flat interval.
Explain why added energy does not raise temperature during that interval.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied latent heat is 200 J/kg.
Find energy needed to change the phase of 3 kg at its transition temperature.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Phase changes

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student uses mcΔT for a melting interval with ΔT=0 and concludes no energy is required.
Explain the missing term.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A spring has k=400 N/m and compression 0.06 m.
Find force magnitude and describe its direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg sample with L=150 J/kg changes phase completely.
Find Q.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2.
Compare heat needed to melt 1 kg and 3 kg of the same substance.
Explain why temperature can remain unchanged in both.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3